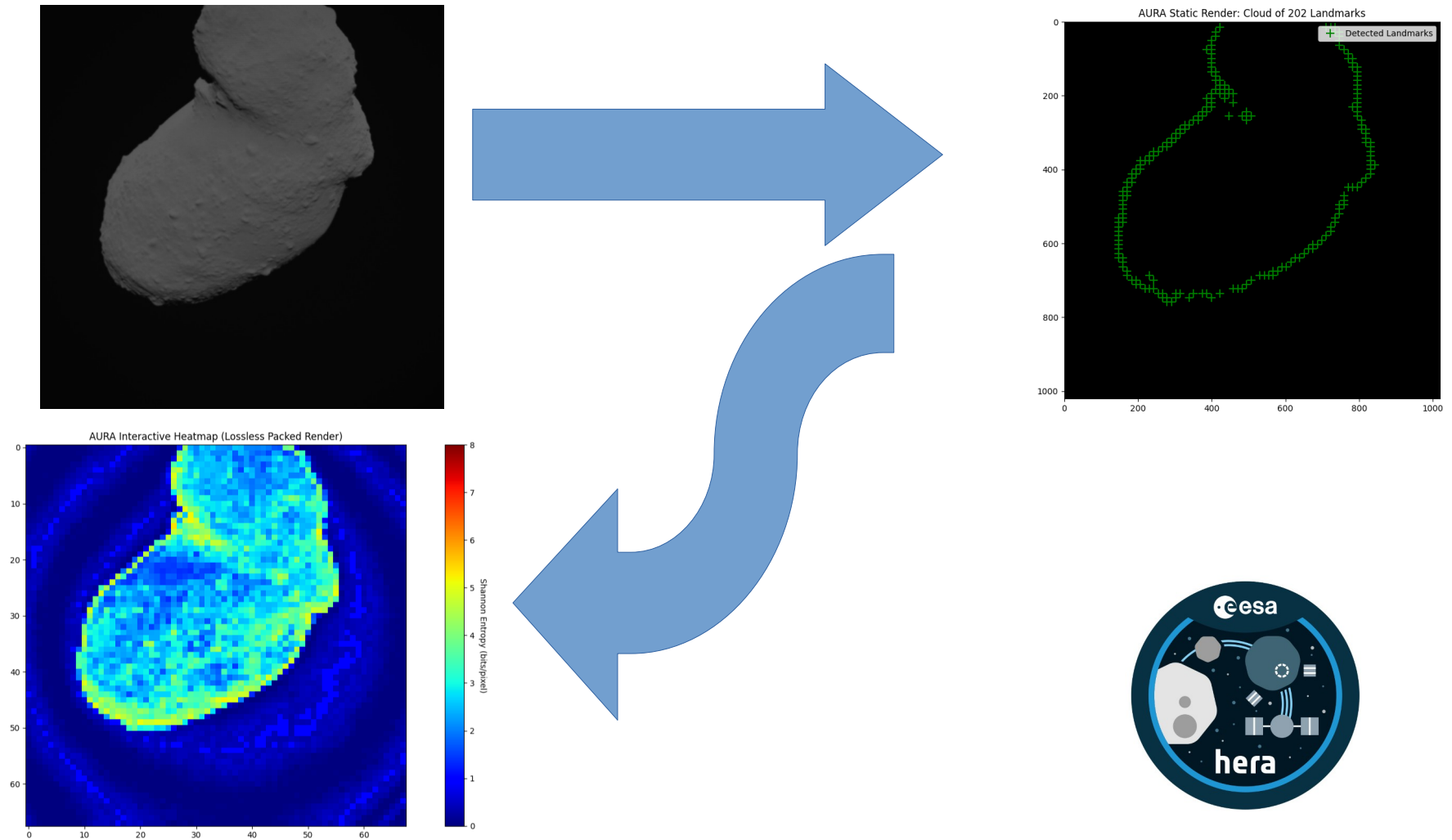


## AURA: Autonomous Unsupervised Feature-Tracking for Real-Time Deep-Space Navigation



The image of the sky body was provided as part of the ESA program - The Open Space Innovation Platform (OSIP).

## Who Am I?

### Serhii Trush Embedded Systems Engineer | Electronics Developer

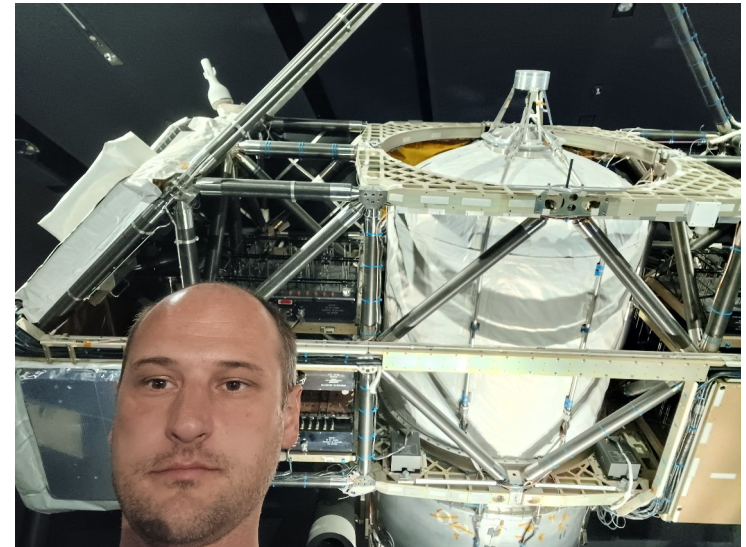
- GitHub: <https://github.com/techn0man1ac/>
- LinkedIn: <https://www.linkedin.com/in/techn0man1ac/>
- YouTube: <https://www.youtube.com/@Techn0man1ac>

### Education

- Computer Systems Engineer(Bachelor degree)
- Engineering background(Commercial / Industrial electronics)
- Electronics and embedded systems
- Software development

### Experience

- 10+ years in electronics
- Embedded development
- Hardware design
- Software engineering



I'm standing next to the CRISTA-SPAS at the Deutsches Museum.



## Origin of the Name “AURA”

The name resonated because it reflects:

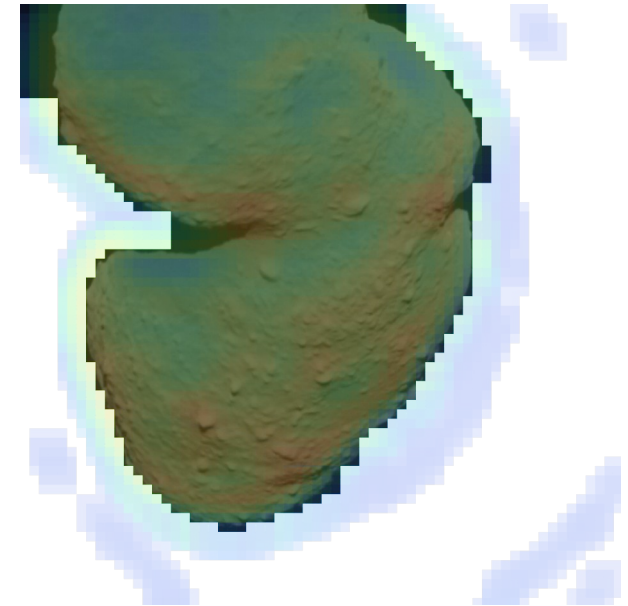
- Information awareness
- Hidden structure detection
- Perception of the environment
- Emergent intelligence

### Philosophical Perspective

An "aura" can be interpreted as: The informational footprint of an object or environment.

### Connection to the Project

AURA attempts to identify and prioritize meaningful information inside large sensor datasets.



Removing image blocks with little information



## Why Hera?

### ESA Hera Mission

- Asteroid exploration
- Planetary defense
- Autonomous spacecraft operations

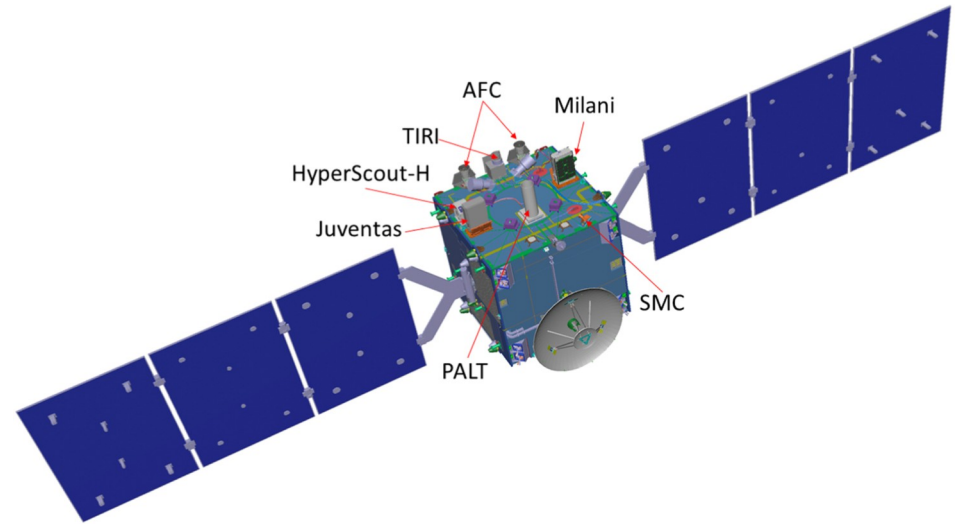
Hera follows the **Open Science philosophy**.

The mission encourages:

- Scientific collaboration
- Transparency
- Publicly accessible research

### Personal Motivation

AURA was inspired by challenges faced by deep-space missions such as Hera.



(c) Hera (space mission)



## The Communication Problem

### Deep Space Constraints

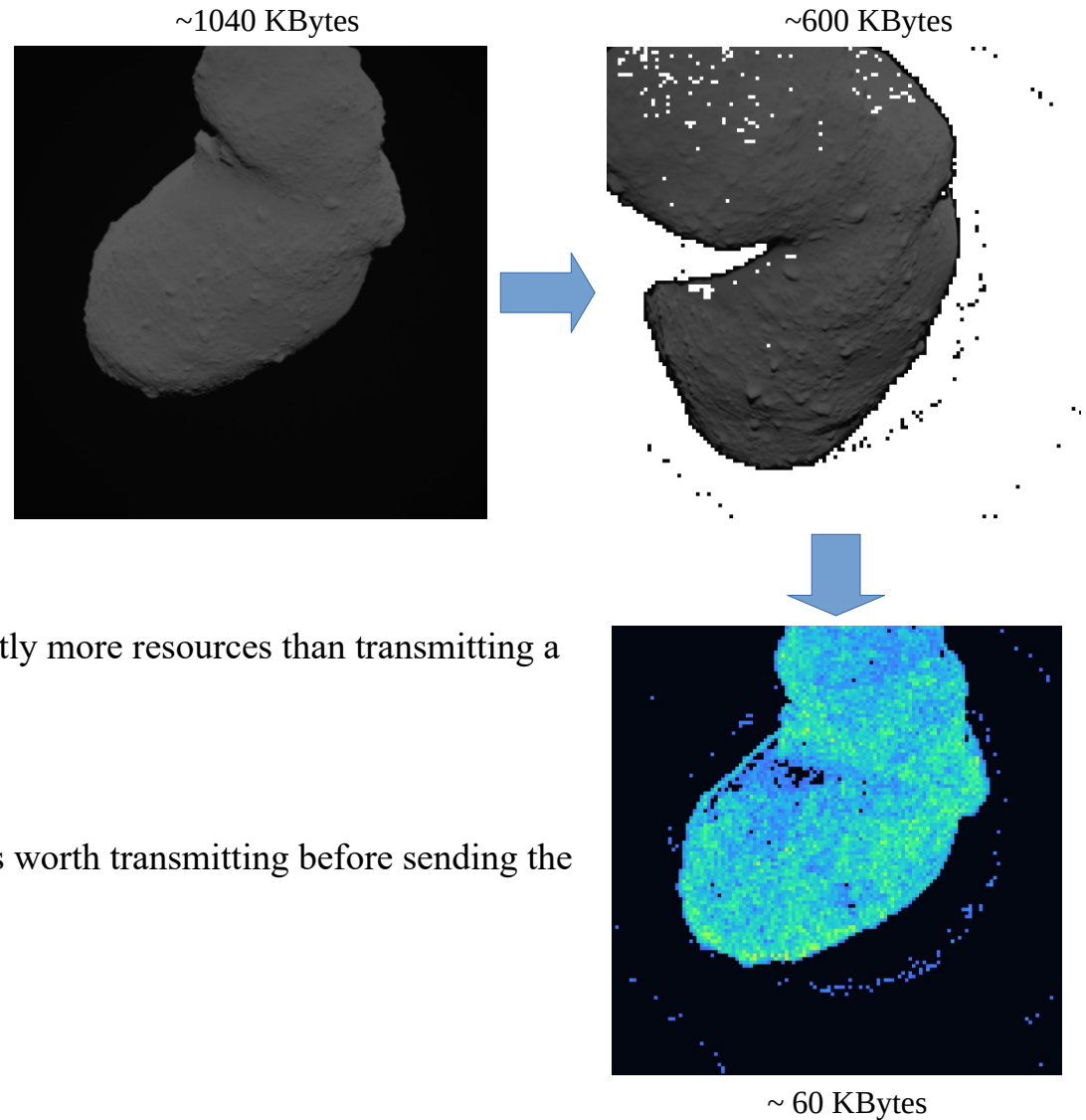
- Long communication delays
- Limited bandwidth
- Limited onboard computing resources
- Power constraints

### Example

Transmitting a full image may require significantly more resources than transmitting a compact information summary.

### Research Question

Can a spacecraft decide what information is worth transmitting before sending the full image?



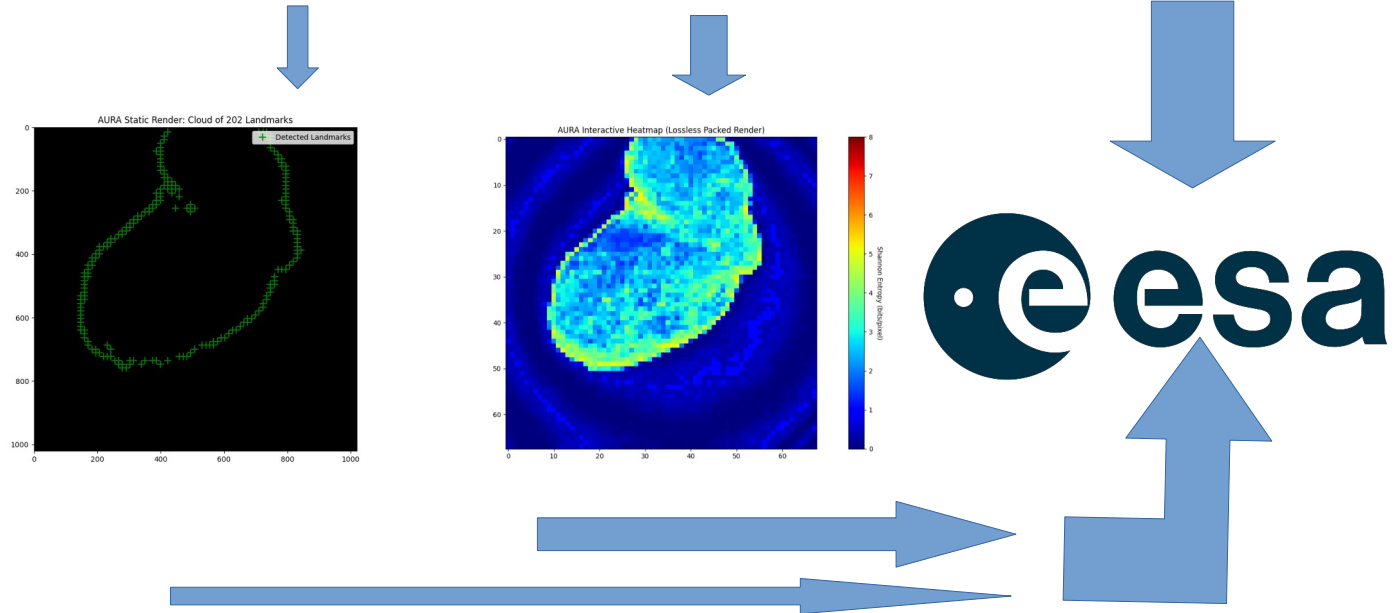
## Core Idea

Can pixels control everything? Can entropy take responsibility?

Not all pixels carry the same amount of information.

## AURA Workflow

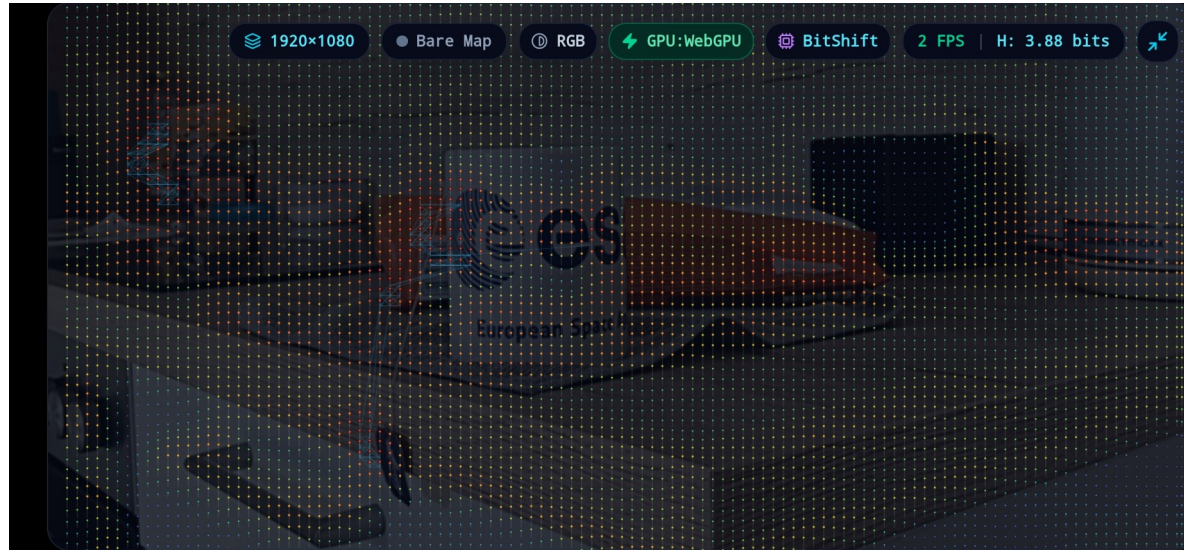
Cameras image → Information Analysis → Feature Point Cloud → Entropy Map → Decision → Data Transmission



## AURA goal:

- Reduce unnecessary data transmission while preserving scientific value.

## Google AI Studio [Sandbox](#)



## Purpose?



## Why It Matters?

Fast experimentation environment for:

- Entropy visualization
- Feature extraction
- Point cloud generation
- Constellation generation
- Concept demonstration

Allows rapid iteration before implementing ideas on embedded hardware.

- Real-time entropy mapping
- Point cloud(Constellation visualization)
- Quick idea validation

## Technical Details

### Image Processing

- Local entropy calculation
- Contrast maxima detection
- Sparse feature extraction

### Embedded Optimizations

- Bit-wise operations
- Memory-aware design
- LEON3 compatibility
- Low computational overhead

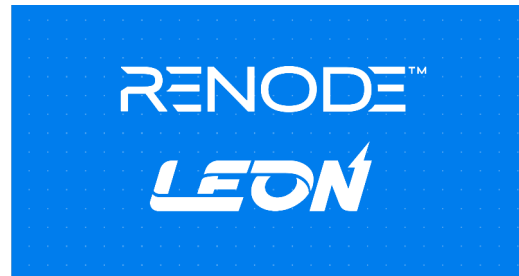
### Current Status

- Emulator implementation
- Functional prototype
- Experimental visualizations

```
PS C:\Projects\bcc-2.2.3-gcc-mingw64\leon3_project_v_0_2> renode .\script.resc
17:21:00.9674 [INFO] Loaded monitor commands from: C:\Program Files\Renode\scripts/monitor.py
17:21:01.6251 [INFO] Including script(s): C:\Projects\bcc-2.2.3-gcc-mingw64\leon3_project_v_0_2\script.resc
17:21:01.6425 [INFO] System bus created.
17:21:02.2423 [INFO] sysbus: Loading block of 10427808 bytes length at 0x40000000.
17:21:02.2960 [INFO] cpu: Setting PC value to 0x40000000.
17:21:02.2968 [INFO] cpu: Using PC value as Entry Point value : 0x40000000
17:21:02.3067 [INFO] sysbus: Loading block of 1040400 bytes length at 0x40600000.
17:21:02.6215 [INFO] leon3-space-sandbox: Machine started.

Renode
RENODE™
Renode, version 1.16.1 (d66b0c2a-202602161036)
(monitor) i $CWD/.\script.resc
Starting emulation...
(leon3-space-sandbox) █
```

LEON3 CPU support in Renode



## Development Road map(up to TRL7)

### Today ~TRL4

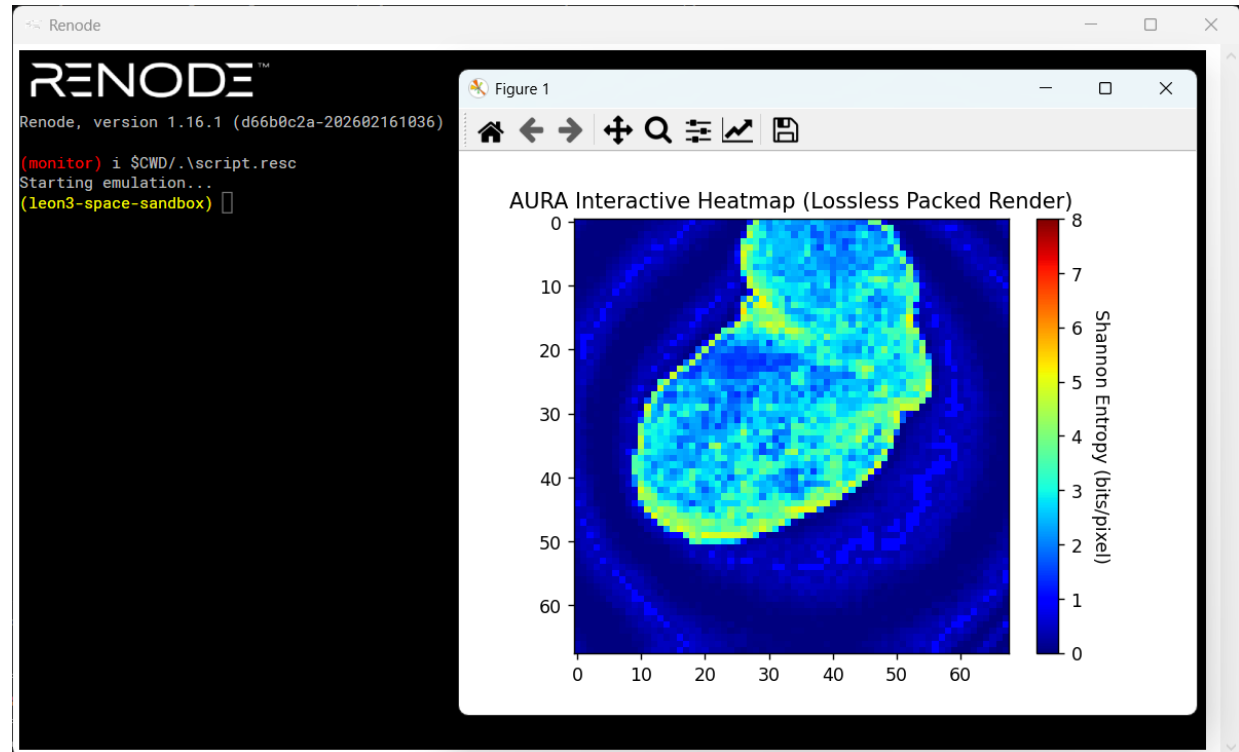
- Concept
- Software prototype
- LEON3 emulation

### Next Steps

- Hardware validation
- Radiation-tolerant platforms
- Integration testing
- Mission-relevant datasets

### Long-Term Goal

TRL progression toward operational space systems.



# Why a PhD?

## Scientific Questions

Can entropy-based representations:

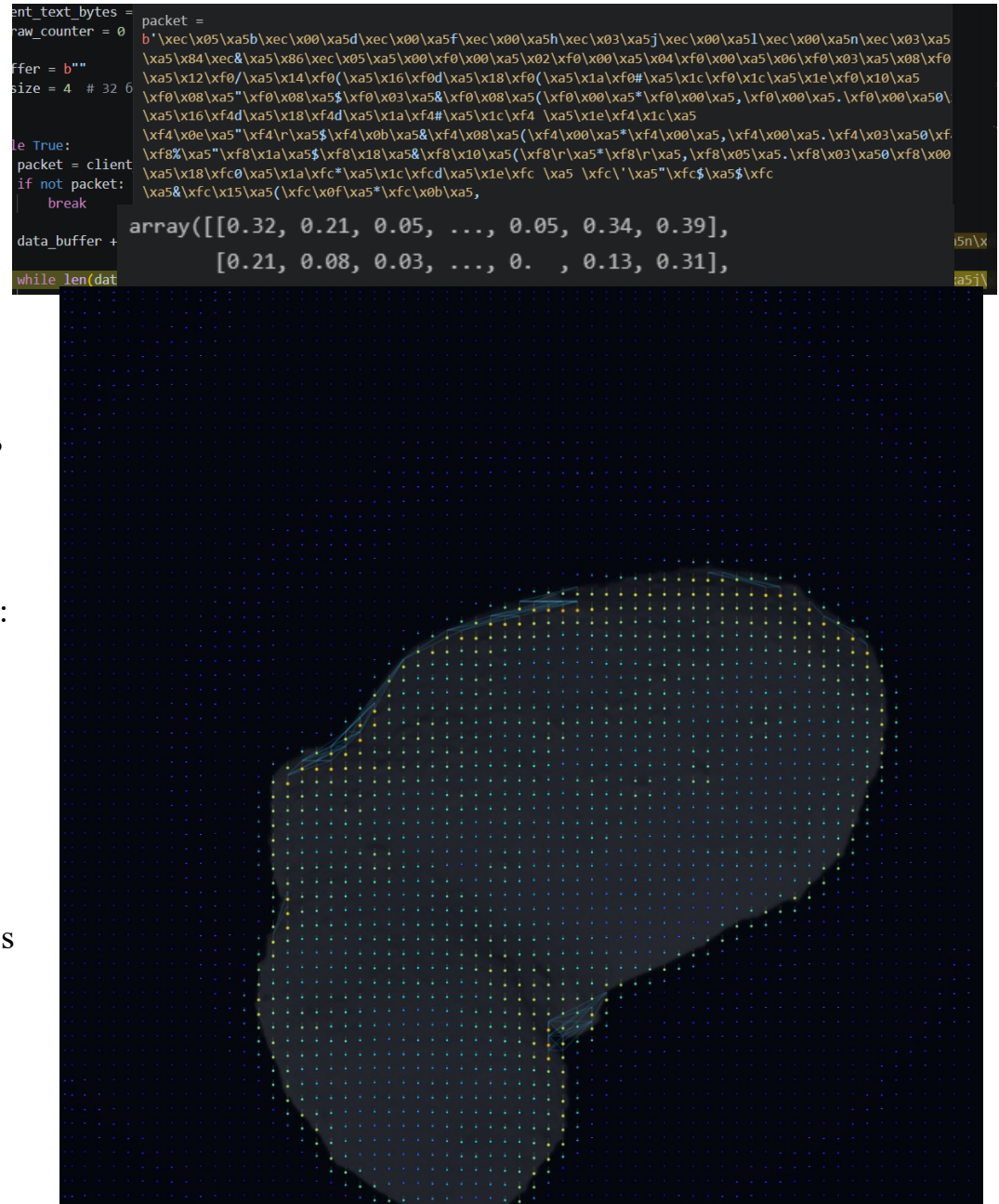
- Can pixels/entropy control a spaceship?
- Reduce bandwidth requirements?
- Improve onboard decision making?
- Support visual navigation?
- Enable multi-sensor information fusion?

## Research Potential

AURA currently opens several research branches:



1. Data prioritization
2. Information compression
3. Visual navigation
4. Joint entropy across sensors



## Beyond Technology

### Personal Goal

Build a long-term academic and professional future in Germany.

### Motivation

- Contribute to German research
- Participate in European space projects
- Collaborate with universities and industry

### Long-Term Vision

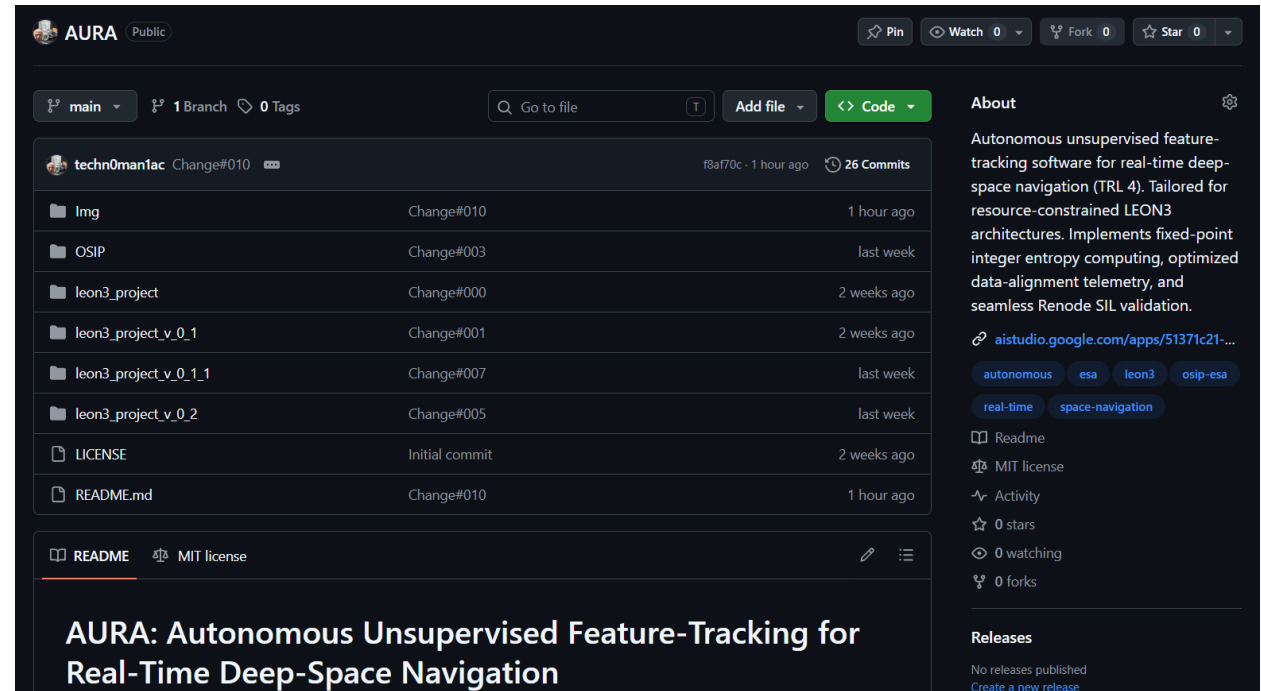
Growing together with the German scientific and engineering community.



# AURA

## Current Status

- Working prototype
- Clear research direction
- Open-source development(MIT license)
- Potential space applications



<https://github.com/techn0man1ac/AURA/>

## Key Message

AURA explores how information itself can become a resource for autonomous space systems.

**Thank You**

